

Scalability & Future Plan

Date	30 th September 2026
Team ID	SWTID-2026-2214
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Access is currently configured mainly for the EEE branch.	Access is currently configured mainly for the EEE branch.	Medium
2	ACL rules are configured manually in ServiceNow.	Changes may require administrator configuration.	Medium
3	Testing is performed using selected users and branch records.	More extensive testing is needed for larger numbers of users and records.	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Access is tested with a limited number of users.	Extend role-based access control to support more users and departments.
2	Data Storage	Institution records are stored in the u_institution_details table.	Add more fields and records as the institution grows.
3	Performance	ACL scripts check user roles and branch values during access.	Optimize ACL scripts and conditions for larger data volumes.
4	Security	Read, Create, Write and Delete operations are controlled using ACLs.	Add more field-level and role-based security rules for stronger access control.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Extend ACL rules to additional branches such as ECE and CSE.	1–2 Months	Supports access control for more departments.
Phase 2	Add more roles and user groups for different access levels.	2–3 Months	Provides flexible role-based access.
Phase 3	Add detailed access monitoring and logging.	3–4 Months	Improves security tracking and auditing.
Phase 4	Extend the solution with more advanced field-level access controls.	4–6 Months	Provides stronger and more detailed record protection.